Software Project Management 2025/2026 – Lab Classes
Licenciatura em Engenharia Informática

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1. Project Assignment

“Software Project Management”

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1. Goal

- The goal of this project is to build a software product following a well-defined development plan and quality assurance plan, within the expected scope, quality, budget and deadline.

2. Theme

- The product to be built shall be simple enough that a single person could program it in a couple of days.
- It should be original work.
- It should be built in java language.
- The choice of the theme is up to the team; however, it must be approved by the teacher.

3. Groups

- Group size shall be, ideally, of 5 members (can be 4).
- All group members must be proficient in the Java programming language.
- There are 4 roles:
  - Scrum Master (SM) - guides the team in applying Scrum, fosters collaboration, and removes impediments to maximize productivity
  - Product Owner (PO) - defines and prioritizes requirements, ensuring the team delivers the most valuable features aligned with product vision and stakeholder needs
  - Developers (Dev) - are all the team members that create the product, by planning, designing, coding, testing, ensuring quality, adapting work, and holding each other accountable
  - Quality Assurance Engineer (QA) – not a formal role from the Scrum Guide – ensures user stories meet requirements, validates completion, and manages merging into the main branch
- Each member has one main role and may have a secondary role
- The main roles change every sprint

4. Project repository

- Each team shall maintain a GitLab repository to manage
  - A README with all documentation
  - Backlogs
  - Code
  - Sprints
  - Project status
  - Logs
  - …
- The teams use the Assignments in MS Teams to deliver
  - A PPTX that supports each ceremony
  - The final presentation

5. Project Lifecycle

- Create Team (1 week)
  - Team
  - Initial project idea
- Project Setup (3 weeks)
  - Vision & Scope
  - Setup working environment
- Sprint 0 (2 weeks)
  - Project Planning
- Sprint1+2 (2+2 weeks)
  - Release 1 - MVP
- Sprint3+4 (2+2 weeks)
  - Release 2 - Final
- Closeup (1 week)
  - Final report

6. Project assessment

- At every ceremony, the team is evaluated:
  - Week 5 – KOM (20%)
  - Weeks 7, 8, 11, 13, 15 – Sprint review (5x12%=60%)
  - Week 15 – Final presentation (20%)
- Every other week, the team receives feedback

- Scores are public and compared
- Teams compete between each other!

7. Individual Evaluation

- Individual grades start from group grades, adjusted by an individual factor
  - based on peer assessment within the team
  - based on the student’s performance during the semester (participation in forums, GitLab, meetings, etc.)

- Mid-semester peer assessment within the team is encouraged
- Students are encouraged to **assess the teammates** in the middle of the semester
- Students will assess the teammates at the end of the semester

8. Work effort

- **4 *hours per person-week* (out of class)**
- 4 x number of team members *hours per team-week* (ex: 4x5=20 *hours per team-week*)
- 4 x number of team members *hours per sprint* (ex: 4x5x2=40 *hours per sprint*) = Sprint capacity

- The real effort must be logged

9. Student Responsibilities

- Students are required continuous and regular work throughout the semester
- Students must ensure availability, act responsibly, and be proactive
- Groups may request the exclusion of a member due to repeated lack of participation, with teacher's approval

- A **Learning Agreement** must be accepted by every student – see MS Teams

10. Responsible Use of Artificial Intelligence

- AI tools may be used to support project documentation
  - Examples: Vision & Scope, User Stories
  - Usage must be explicitly referenced
- AI in coding and testing is not only allowed, but encouraged
  - Helps students experience practices widely adopted in the industry
- **Students are fully responsible** for all documentation and code they produce
  - Responses such as “the bot told me” or “I don’t remember” are **heavily penalized**

Good luck with your software project!

2. GitLab

README.md document

1. Team
2. Vision and Scope
3. Requirements
    1. Use case diagram
    2. Mockups
    3. User Stories (links to GitLab issues)
4. DoD (Definition of Done)
5. Architecture and Design
    1. Domain Model
    2. Sequence Diagram
6. Risk Plan
7. Pre-Game
8. Release Plan
    1. Release 1
    2. Release 2
9. Increments
    1. Sprint 1
    2. Sprint 2
    3. Sprint 3
    4. Sprint 4

1. Team

- Team members:
  - Name1, student number, GitLab email
  - Name2, student number, GitLab email
  - Name3, student number, GitLab email
  - …

2. Vision and Scope

1 Problem Statement

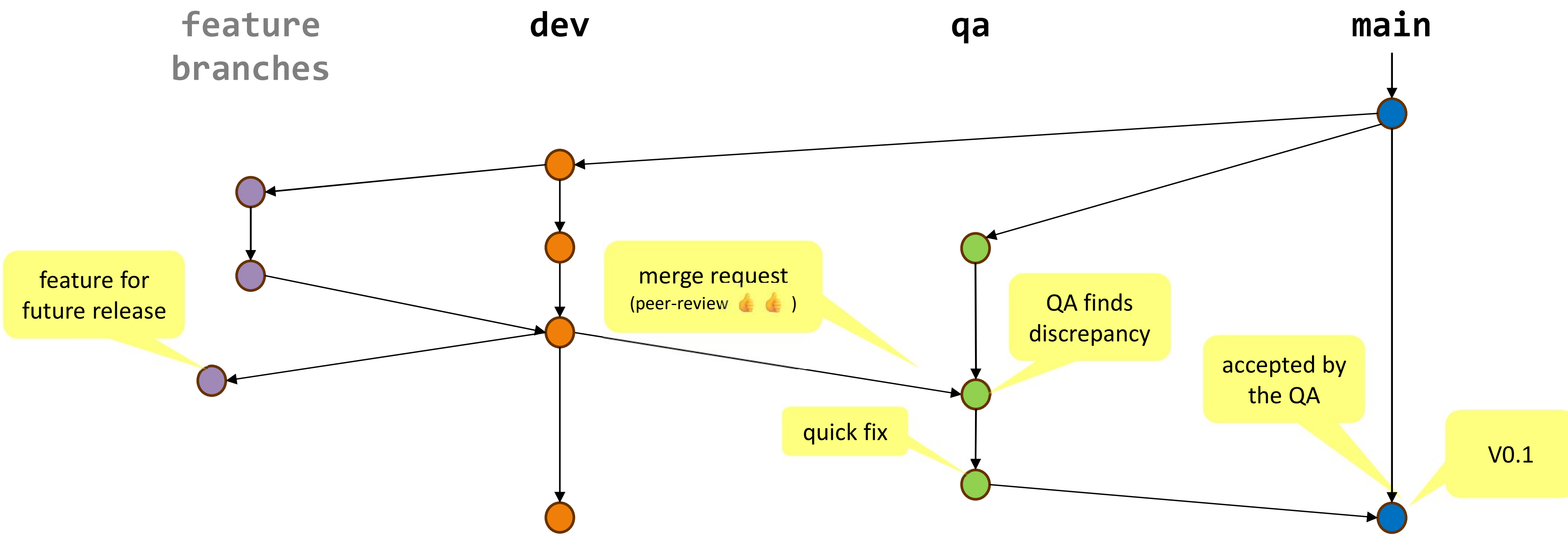
- 1.1 Project background
- 1.2 Stakeholders
- 1.3 Users

2 Vision & Scope of the Solution

- 2.1 Vision statement
- 2.2 List of features
- 2.3 Features that will not be developed
- 2.4 Assumptions

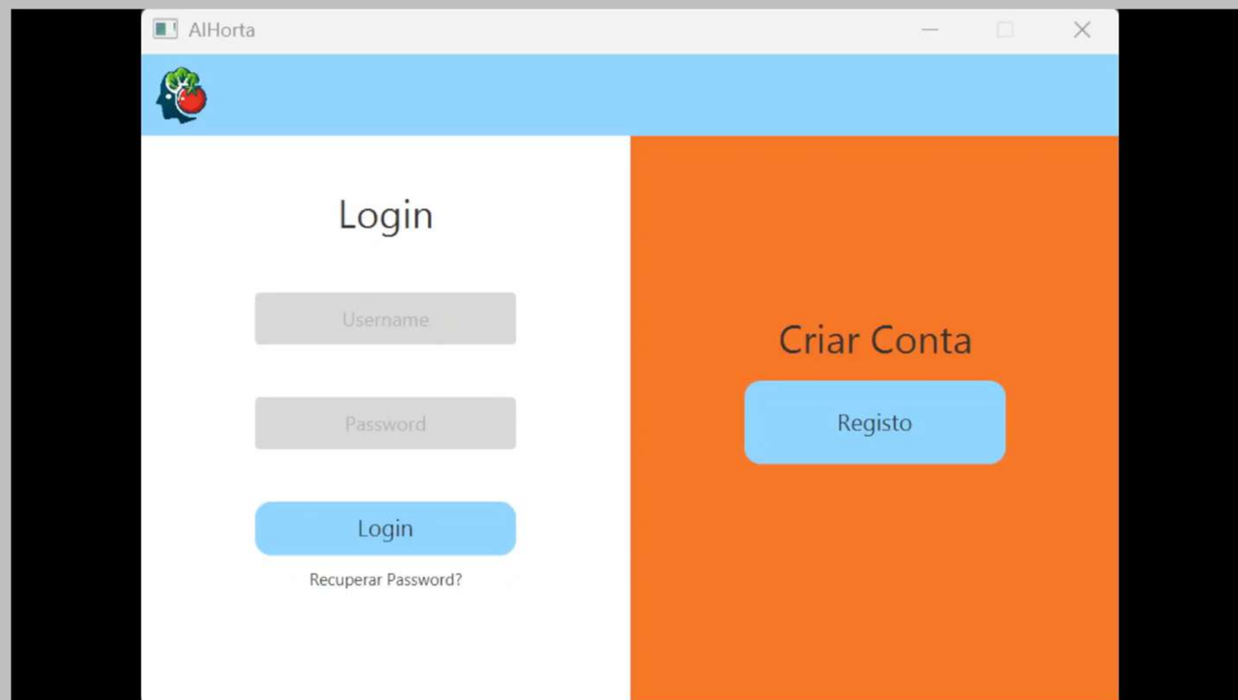
GPS Git workflow

- Branches



Example from last year

- AiHorta



TO DO for Week 3 – baseline tools

1. Install IntelliJ (https://www.jetbrains.com/help/idea/installation-guide.html)
2. Install and configure GitLab
   - Create GitLab account
   - Join the Educational Program
   - Create your team
     - Name: GPS2526_GroupXY (X - Lab number; Y team number)
       - Ex: GPS2526_Group11
     - Add teachers to the team
     - All members as owners
   - Update README
     - 2. V&S
     - 3.1+3.2 Requirements